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Real-Time Road Sign Localisation through Object Detection

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Abstract

The detection of road signs is pivotal for effective traffic management and the operation of autonomous vehicles. In this context, deep learning, especially with respect to object detection models, can make a substantial contribution by facilitating the direct detection of road signs from images captured by moving vehicles. In this paper, we propose a method aimed at detecting road signs by exploiting the You Only Look Once model. We experiment with a dataset composed by 10000 different road sign images by obtaining interesting performances, thus showing that the proposed model can be considered reliable for real-time road sign detection.

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1. Introduction and Related Work

Road sign detection is indeed important and has several practical applications, especially in the context of road safety, autonomous vehicles, and intelligent transportation systems. Here are some reasons why road sign detection is considered significant:

- **Traffic Safety:** Road signs are crucial for conveying important information to drivers, such as speed limits, warnings, and regulatory instructions. Accurate detection of road signs helps ensure that drivers are aware of and comply with the rules and regulations, contributing to overall road safety.
- **Driver Assistance Systems:** Advanced Driver Assistance Systems rely on road sign detection to provide real-time information to drivers. These systems can warn drivers about speed limits, upcoming stop signs, and other relevant information, enhancing driver awareness and reducing the risk of accidents.

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- *Autonomous Vehicles*: Autonomous vehicles heavily depend on computer vision and sensor technologies for navigation. Road sign detection is a critical component of autonomous driving systems, enabling vehicles to interpret and respond to traffic signs, signals, and road markings.
- *Navigation Systems*: Navigation systems in modern vehicles use road sign information to provide accurate directions and guidance to drivers. Detecting and interpreting road signs helps navigation systems offer timely and relevant information to the driver.
- *Efficient Transportation Systems*: Efficient transportation systems benefit from the accurate detection of road signs. This includes the optimization of traffic signal timings, the management of intersections, and the implementation of intelligent transportation solutions to improve overall traffic efficiency.
- *Smart Cities*: In the context of smart cities, road sign detection contributes to the development of intelligent transportation infrastructure. It enables the integration of data from road signs into broader smart city systems for improved urban planning and management.

As highlighted, road sign detection is a fundamental aspect of modern transportation systems, contributing to road safety, driver assistance, and the development of intelligent and efficient transportation solutions. Advances in computer vision, machine learning, and sensor technologies continue to enhance the accuracy and reliability of road sign detection systems.

Computer vision is a field of study that enables machines to interpret and make decisions based on visual data from the world. It involves the development of algorithms and techniques to enable machines to understand and extract information from images or videos. The ultimate goal of computer vision is to enable machines to perform tasks that typically require human vision, such as image recognition, object detection, and scene understanding.

Object detection is a specific task within computer vision that focuses on locating and classifying objects within an image or video. The goal is to identify the presence of objects in an image and determine their locations by drawing bounding boxes around them. Object detection goes beyond simple image classification, where the task is to assign a label to the entire image. Instead, object detection involves identifying and classifying multiple objects within an image.

There are various approaches to object detection, including traditional computer vision techniques and more recent deep learning methods [1, 17, 11]. Deep learning, particularly convolutional neural networks [18, 2, 7, 3], has proven to be highly effective in object detection tasks. One of the most popular object detection frameworks is You Only Look Once (YOLO)[8].

Starting from these considerations, in this paper we propose a method to detect road signs directly from images. In particular, we aim to demonstrate that it is possible to perform road sign detection directly from images by considering the YOLO model version 8.

Researchers in [13] already proposed the adoption of the YOLO model in this context, in particular, they adopted the version 5 of the YOLO model for road sign detection. They obtained a precision equal to 0.881, while the proposed model is able to reach a precision equal to 0.971 (thus showing better performances), as discussed in the experimental results section.

The following of the paper proceeds as follows: in the next section, the proposed method is presented, in Section 3 the results of the experimental evaluations, conducted by employing a dataset composed by 10000 different road sign images, are discussed and, finally, in the last section conclusion and future research directions are drawn.

2. A Method for Real-Time Road Sign Detection

This section introduces the proposed method for detecting and localizing road signs.

Specifically, our method is designed to automate road sign classification, localization, and detection percentage directly from images, such as those captured by moving vehicles, through the application of deep learning techniques.

Figure 1 illustrates the proposed approach, where the method not only identifies road signs but also provides the prediction percentage for each detection. This percentage serves as a measure of the confidence of the model in the accuracy of a given prediction.

To develop an effective deep learning model for object detection in images captured from moving vehicles, a dataset comprising images of road signs is essential. Specifically, images obtained from various distances are needed.

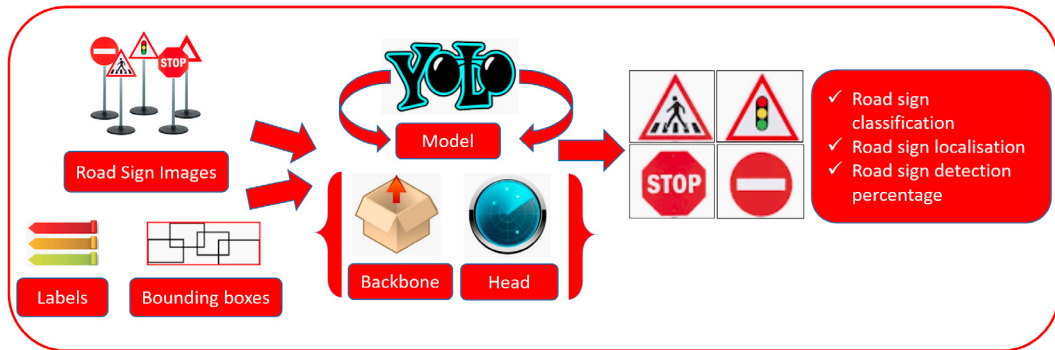


Fig. 1. The proposed method.

To construct a model capable of not only recognizing the presence of objects in an image but also pinpointing their location, a set of images detailing the object's position is required. Consequently, the images undergo manual labeling and annotation to mark the areas in the analyzed images where the objects are present, as illustrated by the bounding box in Figure 1. The classes (i.e., *classes* in Figure 1) for the detection of the bounding box are the following: Road narrows on right, 50 mph speed limit, Attention Please, Beware of children, Cycle Route Ahead Warning, Dangerous Left Curve Ahead, Dangerous Right Curve Ahead, End of all speed and passing limits, Give Way, Go Straight or Turn Right, Go straight or turn left, Keep-Left, Keep-Right, Left Zig Zag Traffic, No Entry, No-Over-Taking, Overtaking by trucks is prohibited, Pedestrian Crossing, Round-About, Slippery Road Ahead, Speed Limit 20 KMPH, Speed Limit 30 KMPH, Stop_Sign, Straight Ahead Only, Traffic_signal, Truck traffic is prohibited, Turn left ahead, Turn right ahead, Uneven Road. We contextualize the proposed model to the detection of these 29 different classes but it can be applied also to other detection contexts.

Furthermore, to construct a model that is not only efficient but also capable of accurately predicting unseen images, the images were captured from various angles and under different conditions. All the images have varying sizes, but after obtaining them, a preprocessing step is necessary to resize the images to a uniform dimension.

After acquiring the dataset, the next step involves selecting a model for training. In this study, we opt for the YOLO model. YOLO, introduced by J. Redmon et al. in 2016 [14], represents the pioneering one-stage deep learning detector. YOLO functions as an object detection model with the dual objectives of classifying images and accurately determining the positioning of objects within them [10]. Specifically, we utilize version 8 of the YOLO model, more precisely, the yolov8s version [6].

The primary distinction from other models lies in YOLO's approach, which employs an independent pipeline for the entire process [9]. In this approach, each analyzed image is partitioned into a matrix of $S \times S$ cells, with a cell being designated for the object if the object's center falls within the cell [12, 8].

In contrast to existing object detection models, YOLO exhibits notable speed advantages, as evidenced in studies such as [16, 15].

This efficiency is primarily attributed to YOLO's unique approach, as it does not partition the recognition process into multiple phases. Instead, YOLO predicts bounding boxes, probabilities, and object classes in a single phase for the objects present in the input image.

We consider this model because, in comparison to other deep learning models for object detection, YOLO has certain advantages. Despite being acknowledged for potentially more localization errors [5], YOLO is simultaneously less prone to recognizing false positives in the background of the image. Additionally, it is significantly faster [4, 8]. These factors collectively contribute to YOLO being widely regarded as one of the best convolutional neural network models for object detection. We implement the model with the PyTorch¹ framework the YOLOv8s² version of the YOLO model.

¹ <https://pytorch.org/>

² <https://www.ultralytics.com/>

The YOLO network is composed of a backbone, specifically a convolutional neural network that consolidates and structures image features at various granularities. Additionally, it includes a Head designed to utilize features from the neck and perform box and class prediction steps. Situated between the backbone and the head is the neck, which consists of a series of layers intended to blend and integrate image features before forwarding them for prediction.

3. Experimental Analysis

This section showcases the outcomes of our experimental analysis, demonstrating the efficacy of the YOLO model in road sign detection.

We obtained the road sign images from the Traffic and Road Signs Computer Vision Project dataset, specifically designed for constructing models capable of directly detecting road signs from images. The dataset is freely available for research purposes³.

This dataset is comprised of many different road signs, it is composed of 10000 images with 10000 annotations (1 annotation per image, across the following 29 classes: Road narrows on right, 50 mph speed limit, Attention Please, Beware of children, CYCLE ROUTE AHEAD WARNING, Dangerous Left Curve Ahead, Dangerous Right Curve Ahead, End of all speed and passing limits, Give Way, Go Straight or Turn Right, Go straight or turn left, Keep-Left, Keep-Right, Left Zig Zag Traffic, No Entry, No_Over_Taking, Overtaking by trucks is prohibited, Pedestrian Crossing, Round-About, Slippery Road Ahead, Speed Limit 20 KMPH, Speed Limit 30 KMPH, Stop_Sign, Straight Ahead Only, Traffic_signal, Truck traffic is prohibited, Turn left ahead, Turn right ahead, Uneven Road. To ensure the replicability of results, we opted to utilize a publicly available dataset. The images were taken at different distances and are stored in the jpg file format with a resolution of 416 x 416 pixels.

The dataset was partitioned as follows: 7092 images for training, 1884 for validation, and the remaining 1024 for testing. No augmentations were applied during this process.

The acquired dataset is annotated, meaning each image includes detailed information about the bounding box around each road sign.

Concerning the model parameters, we chose a batch size of 16, set the number of epochs to 25, and initialized the learning rate at 0.01.

For model training, we utilized a machine equipped with an NVIDIA Tesla T4 GPU card with 16 GB of memory.

The model training duration was 1.949 hours, with this value corresponding to the wall clock time for 25 epochs.

Figures 2 present the experimental results obtained by the proposed method through multiple plots. In the first row of plots in Figure 2, there are the following: train/box_loss (indicating the trend of box_loss during training, a measure of how closely the predicted bounding boxes align with the ground truth object), train/obj_loss (representing the trend of obj_loss during training, where objectness determines the presence of an object at an anchor), train/cls_loss (illustrating the trend of cls_loss during training, a measure of the correctness in classifying each predicted bounding box, considering that each box may contain an object class or background; commonly referred to as cross-entropy loss), precision trend, and recall trend. In the second row of plots in Figure 2, there are: val/box_loss (depicting the trend of box_loss in validation), val/obj_loss (showing the trend of obj_loss in validation), mean Average Precision when Intersection over Union is equal to 0.5 (mAP_0.5), and mean Average Precision when Intersection over Union is between 0.5 and 0.95 (mAP_0.5:0.95).

All the metrics demonstrate the anticipated trends: indeed, precision, recall, mAP_0.5, and mAP_0.5:0.95 are expected to exhibit a growth trend with increasing epochs, indicating the model's correct learning of object detection from road signs. Conversely, the other metrics display an opposing trend, showing a decrease as the epoch number increases. This serves as a secondary confirmation that the model is effectively learning from road sign images. The loss metrics, which generally highlight instances when a model incorrectly identifies a specific object, are typically very high in the initial epochs but are expected to decrease with the increase of epochs as the model becomes adept at detecting objects of interest.

Below, additional details are furnished regarding the precision, recall, mAP_0.5, and mAP_0.5:0.95 metrics.

³ <https://universe.roboflow.com/usmanchaudhry622-gmail-com/traffic-and-road-signs>

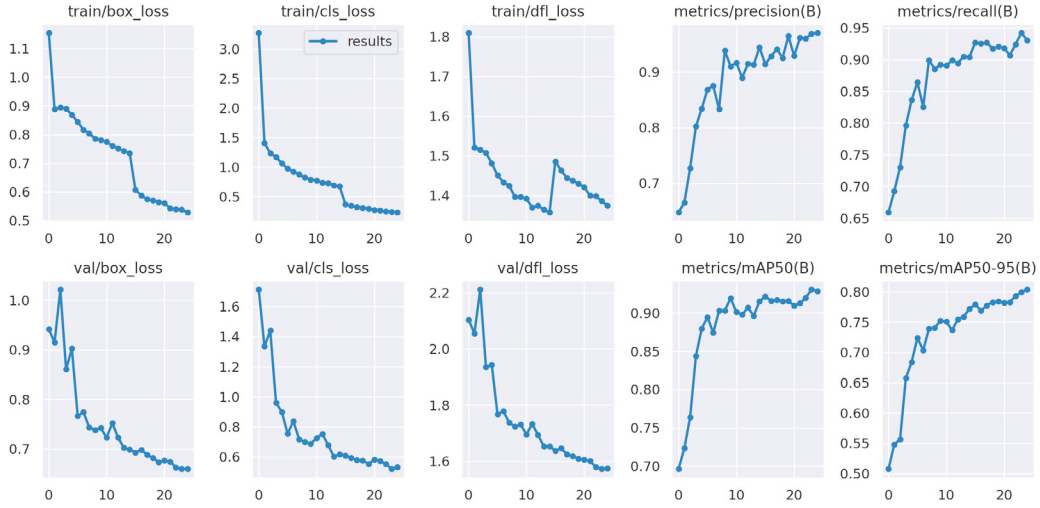


Fig. 2. Experimental analysis results.

Precision measures the proportion of positive predictions that are accurate. It considers false positives i.e., the instances incorrectly identified for inclusion. Precision can be computed as:

$$Precision = \frac{TP}{TP+FP}$$

Observing the precision trend in Figure 2, it is evident that this trend increases across various epochs until it stabilizes, notably around the 20th epoch. This pattern indicates that the network effectively learns to differentiate between the presence and absence of road signs throughout the different epochs.

The second metric we calculate to assess the effectiveness of the proposed method is recall. The underlying concept behind this metric is to determine the proportion of actual positives that were accurately predicted. It considers false negatives, representing instances that should have been identified for inclusion but were not. Recall can be computed as:

$$Recall = \frac{TP}{TP+FN}$$

Examining the recall trend illustrated in Figure 2, a behavior akin to the one highlighted for precision becomes apparent. Indeed, both precision and recall should exhibit an increasing trend with the growing number of epochs. This observed trend is reflected in the plots of Figure 2 for both metrics, and given that precision and recall values range from 0 to 1, the achieved performances are promising. Similar to precision, recall also demonstrates an increasing trend with the augmentation of epochs.

Precision and recall are commonly used metrics to assess the performance of a model in classification tasks. Clearly, to evaluate whether the model can accurately localize the object of interest in road sign images within the correct region of the analyzed image, certain metrics are required. One such metric is the AP (Average Precision), a classic metric dedicated to measuring the accuracy of object detectors, such as the YOLO model we utilized. Average precision calculates the average precision value for recall values ranging from 0 to 1.

We focus on computing the mean Average Precision (mAP), a metric that involves Intersection over Union (IOU), Precision, Recall, Precision-Recall Curve, and AP. In object detection models, predictions encompass both the bounding box and the category of objects within an image. IOU is employed to assess the accuracy of the bounding box prediction.

The IOU signifies the degree of overlap between bounding boxes. This ratio of overlap ranges from 1.0 in the case of an exact match to 0.0 when there is no overlap between the regions of two bounding boxes. The IOU formula is shown below:

$$IOU = \frac{ao}{au}$$

where ao is the area of overlap and au represents the area of union.

In assessing object detection models, it is crucial to establish the threshold for successful recognition based on the overlap of bounding boxes with ground truth data. IOUs serve this purpose, and mAP_{0.5} represents accuracy when

IOU equals 0.5, indicating that detection is deemed successful if there is more than a 50% overlap. A larger IOU implies a higher accuracy requirement for detecting the bounding box, making it more challenging. For instance, the value of mAP_0.75 is lower than that of mAP_0.5.

The mAP is an average of the AP values, which, in turn, is an average of the APs for all classes.

In Figures 2, the plots for metrics/mAP_0.5 and metrics/mAP_0.5:0.95 respectively display the mAP value for IOU=50 and IOU ranging from 50 to 95 (representing different IoU thresholds from 0.5 to 0.95, with a step size of 0.05) on the average mAP.

We observe that the trends depicted in the plots for metrics/mAP_0.5 and metrics/mAP_0.5:0.95 in Figure 2 both exhibit an increasing pattern. Hence, in the context of locating the road signs, the model demonstrates the capability to learn the specific areas in the image that are crucial for accurately highlighting the detected objects.

Figure 1 presents the values obtained for Precision, Recall, mAP_0.5, and mAP_0.5:0.95 metrics, representing the average values of these metrics across all involved classes.

Table 1. Classification results.

Precision	Recall	mAP_0.5	mAP_0.5:0.95
0.971	0.93	0.929	0.804

From Table 1 we can note that the Precision and the Recall are respectively equal to 0.971 and to 0.93. The average time to detect a road sign is equal to 15.3 ms (on the architecture we exploited for the experimental analysis), thus we can consider the proposed method as a real-time detector. In particular, the 15.3 ms obtained as the average time for the detection are related to the following average contributions: 0.7 ms for image pre-processing, 12.3 ms for model inference, and 2.3 ms for image post-processing.

Furthermore, to better evaluate the proposed method, in Figure 3 we report the confusion matrix.

From the confusion matrix shown in Figure 3 we can note that most of the road sign images are rightly classified, as evidenced by the confusion matrix. Only two classes are not rightly detected i.e., No Over Taking and Speed Limit 30 KMPH. In particular, we noted from the experimental analysis results that several road signs related to the speed limit 30 KMPH are misclassified with the speed limit 50 KMPH label. About the remaining label, we can note from the confusion matrix shown in Figure 3 that the majority of road signs were rightly classified in the right class.

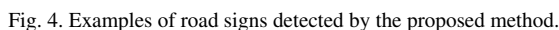
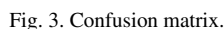
To visually validate the results related to the proposed method and to confirm its effectiveness in a real-world environment in Figure 4 we show two different examples of road signs detected by the trained model.

As shown in Figure 4, Speed Limit 20 KMPH was detected with a percentage equal to 0.9, and a Turn right ahead road signal was detected with a percentage equal to 1.0 (i.e., the percentage varying from 0 to 1 i.e., the maximum value).

From Figure 4 it emerges that the proposed exhibits a good ability in the right localisation of road signs, by demonstrating that the proposed model is capable of recognizing road signs at different distances and in different camera framing (as typically happens when we consider images obtained from moving vehicles that there is no fixed and a priori framing considering that they are vehicles in constant motion).

4. Conclusion and Future Work

Considering the importance of performing real-time and high-accuracy road signs classification and detection, in this paper we proposed a method to detect and localise road signs from images acquired from moving vehicles. We considered the YOLO version 8 model by experimenting with a dataset composed of 10000 different images. We obtained a precision and a recall respectively equal to 0.971 and to 0.93, showing the effectiveness of the proposed model in road sign detection. Moreover, the time to detect a road sign from an unknown image is, on average, equal to 15.3 ms, for this reason, we consider the proposed method able to be applied in a real-time environment. In future work, we plan to consider additional real-time models, with the aim of improving the obtained performances. Moreover, we will consider the proposed YOLO model to detect also stoplights.



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prese e del Made in Italy, Italy, CUP: D33B22000060001 and FORESEEN: FORMal mEthodS for attack dEtEction in autonomous driving systems CUP N.P2022WYAEW.

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